

The Math Behind Chess Engines

There are many games that are played around the world. Some are more popular than others, but there are a few games that, when mentioned, everyone knows what they are even if they don't know how to play them. Soccer, Basketball, Poker, and among this list is chess. I bet if you asked anyone if they know what chess is, they would respond with yes, even if they don't know how to play the wonderful game. But what many don't know is the complexity of chess and the developments made over the years to eventually master chess. Chess itself is filled with a bunch of calculations to decide what is a good move versus what is considered a blunder (a bad move). With chess having these calculations constantly appearing throughout the game, many wondered how computers could possibly help with calculating these moves and even possibly coming up with better moves than the best chess players in the world. That is where chess engines come in.

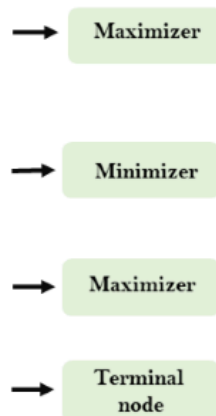
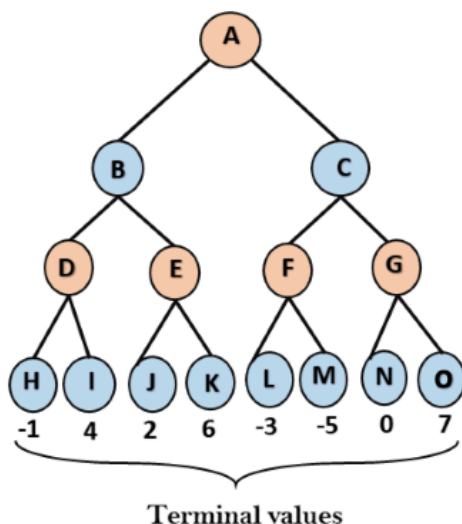
What is a Chess Engine? StockFish?

A chess engine is an advanced computer program designed to analyze and play chess games at a high level. These engines evaluate the positions on the board, calculate possible moves, and determine the best strategies based on various algorithms and data. They have become important tools for both casual players and grandmasters since their creation as they offer insights that were previously unattainable without extensive study and experience.

One of the most well-known and powerful chess engines is Stockfish. Released as an open-source project, Stockfish has undergone continuous development and improvement by a dedicated community of programmers and chess enthusiasts (many people can join in on beta testing and even help provide code). It boasts impressive computational power, capable of evaluating millions of positions per second, and uses advanced algorithms to suggest optimal moves in any position it is tasked with, even if the position is impossible to achieve in a real game. Stockfish's success in various chess competitions and its widespread use in training and analysis make it the most used chess engine in the world as well as a key piece in the chess community as a whole.

MiniMax Algorithm and How Stockfish Applies It

The core of most chess engines is the Minimax algorithm. The Minimax algorithm is a



decision-making algorithm that calculates the worst possible outcome from a decision and tries to minimize the loss. For chess engines, this would mean minimizing the maximum advantage an opponent can have in any given position by analyzing all of the possible moves in

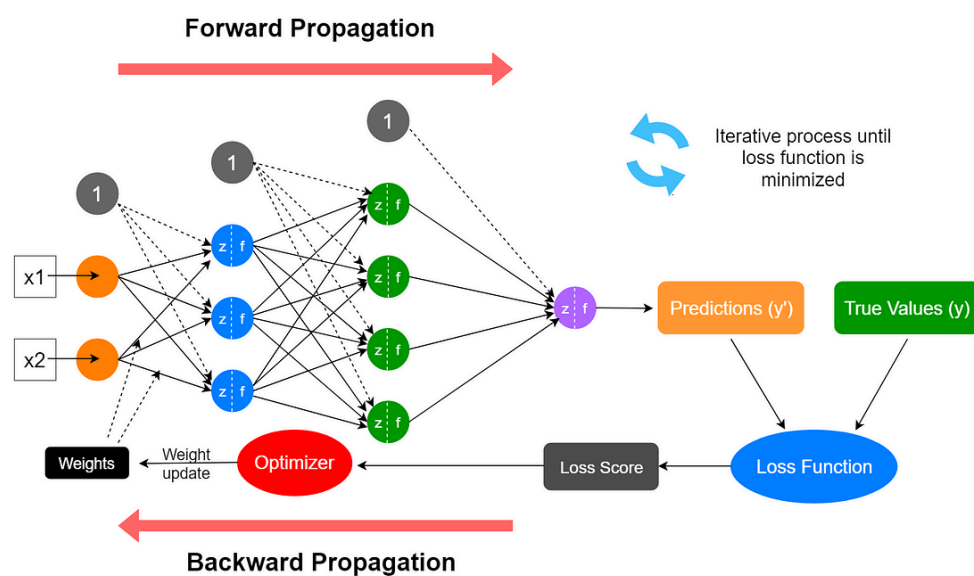
a position, predicting where the opponent would move, and then calculating the move that would be the best for the player.

The problem with the regular Minimax algorithm is that for a game with many possible moves like chess, the time to calculate the best possible move can be very slow and inefficient. This is where Stockfish enhances the Minimax algorithm by using alpha-beta pruning. Alpha-beta pruning reduces the number of nodes necessary for the Minimax algorithm to function, making the algorithm more efficient and thus faster. In terms of the actual chessboard, alpha-beta pruning eliminates moves that would not affect the final decision. With these enhancements, Stockfish is able to perform multilayered analysis that surpasses the capabilities of even the top chess players.

Neural Networks and its Impact on Chess Engines

In recent years, the integration of neural networks into chess engines has revolutionized their performance and capabilities. Neural networks are algorithms that were designed to mimic how a human's brain works. Neural networks consist of layers of interconnected nodes, or "neurons," which process and transmit information. These networks are trained using large datasets, allowing them to learn patterns, recognize features, and make predictions based on the data they have been exposed to.

A neural network typically consists of an input layer, hidden layers, and an output layer. Each neuron in a layer receives input from neurons in the previous layer, it then processes the information from the input layer using an algorithm to pass the result to the neurons in the next layer. This layered structure enables neural networks to model complex relationships and interactions in data, making them highly effective for tasks such as image recognition, language processing, and, in the case of chess, position evaluation and move prediction.



The unexpected rise of AlphaZero (A chess engine developed by DeepMind) showcased the potential of neural networks in chess. AlphaZero uses a deep learning approach, where the network learns and improves by playing millions of games against itself, discovering strategies and patterns without human intervention. This self-learning capability allows neural networks to evaluate chess positions more intuitively and dynamically compared to traditional algorithms which would solely focus on the evaluation of a position through numbers.

Recognizing the advantages of these new neural networks in the chess scene, the developers of Stockfish have introduced Stockfish NNUE. NNUE stands for "Neural Network Update Engine," and it represents a significant enhancement to the traditional Stockfish engine. While Stockfish traditionally relied on handcrafted evaluation functions and brute-force search algorithms, NNUE incorporates neural network evaluations to improve its accuracy and efficiency.

Stockfish NNUE uses a neural network to evaluate chess positions more effectively on top of the algorithms it has used in the past. The network is trained on vast datasets of grandmaster games and high-quality computer-generated games, allowing it to recognize complex patterns and positional nuances that were difficult to encode using traditional evaluation methods, such as multilayered tactics and positions that would just be uncomfortable to play due to a single mistake costing the whole game or positions that have never been played by high level players before. The neural network evaluation is integrated with Stockfish's traditional alpha-beta search algorithm. The engine uses the neural network to evaluate leaf nodes in the search tree, providing more accurate assessments of positions. This hybrid approach combines the learning capabilities of a neural network with the efficient search techniques of traditional algorithms used for chess engines, resulting in a powerful and effective chess engine that based on metrics should outperform anything else . Whilst we have not seen the full capabilities of Stockfish NNUE just yet, many expect it to be able to beat any competition that faces it.

How far Chess Engines Have Come

After learning about the algorithms used to create the chess engines we have today, you may wonder how chess engines performed before our current advancements. Early versions of chess engines were nowhere near as advanced as they are today. They were slow to work and for many years were unable to match the top chess players of their respective times. This all changed in the 1980s, when more research and development were put into them, culminating in 1997 when Deep Blue (IBM's chess engine) beat the World Chess Champion of the time and arguably one of the best chess players of all time, Garry Kasparov. Kasparov was so enraged with the outcome of the match that he claimed the computer cheated on a specific move, alleging it got help from a fellow grandmaster.

Since this remarkable feat, chess engines have only gotten more powerful, to the point that no top chess player can defeat a high-level chess engine. Instead, chess engines are now viewed as tools that can help propel top players even farther ahead of the curve.

Implications of an OpenSource Chess Engine.

With all the algorithms used in creating the strong chess engines today, you would assume that many of these chess engines would be hard to obtain. However, the strongest chess engine right now is an open-source chess engine that anyone can use. Because of this, players and coaches have easy access to chess engines to help them improve by reviewing the games they played. They can see when they made mistakes as well as possible new move combinations they could use in the future. We have seen the skill ceiling for chess rise significantly over the

years, with younger players getting better faster than ever before due to the analysis that these chess engines provide.

The accessibility of powerful chess engines has significantly raised the skill ceiling in the chess world. Players now have tools to deeply analyze positions, study games of top grandmasters(highest title a chess player can achieve), and practice against tough opponents without needing a human partner to play with. This has accelerated the learning curve for many young and upcoming players, enabling them to reach high levels of proficiency at earlier ages. However, this increasing skill ceiling can also be daunting for new players. The complexity and depth of analysis available through these engines may intimidate beginners, making them feel overwhelmed and discouraged from starting such a challenging game.

However, it isn't all positive that these chess engines are so accessible to the public. Since the introduction of chess engines, cheating has become more prevalent than ever, with many top-level players openly discussing other players cheating. We have seen arguably the greatest chess player of all time, Magnus Carlsen, openly refuse to play games against certain opponents due to accusations of cheating. Other past World Chess Champions have also accused many players of cheating. We have even seen current professional chess players banned from certain tournaments due to cheating in the past and possibly even now.

A notable case that highlights the darker side of this accessibility is the controversy involving Hans Niemann(The guy in the selfie who seems uninterested in the chess player standing next to him). In 2022, Niemann, a young(only older than me by a year) and talented

player, faced allegations of cheating during online and over-the-board games. These accusations gained significant attention when Magnus Carlsen, the reigning World Chess Champion, withdrew from a tournament after losing to Niemann, implying potential foul play. The incident sparked a widespread debate about the integrity of competitive chess and the role of chess engines in facilitating cheating. Niemann has denied the accusations, but the controversy has underscored the need for stricter regulations and better detection methods to preserve the fairness of the game.



By making such powerful tools widely available, the chess community faces the challenge of maintaining the integrity of the game. While the benefits of open-source engines are undeniable, the ongoing battle against cheating will require massive efforts from players, organizers, and developers to ensure fair play and to ensure that tournaments are not looked down upon for how they handle situations. The rise in cheating incidents has led to increased scrutiny and the implementation of advanced anti-cheating measures in tournaments. These measures include more sophisticated monitoring technology, stricter regulations, and harsher penalties for those found guilty of cheating. Fun fact, on top of the metal detectors top players usually have to face before matches, since these cheating scandals, they have held the metal detector closer to possible areas a cheating device could be. (Many accused Hans of possibly using a butt plug with

a chess engine in it to help with his games)



Despite these challenges, the accessibility of powerful chess engines continues to contribute positively to the chess world. It continues to allow easy access to high-level chess knowledge, allowing players from all backgrounds to improve their skills. However, as the skill ceiling continues to rise and the complexity of the game becomes more apparent, the chess community must find ways to make the game more approachable for new players. Encouraging a culture of learning and support, along with providing resources for beginners, can help mitigate the intimidation factor and foster a more inclusive environment for all players.

Where pictures are from:

MiniMax Algorithm Picture: <https://www.javatpoint.com/mini-max-algorithm-in-ai>

Neural Network Picture:

<https://medium.com/data-science-365/overview-of-a-neural-networks-learning-process-61690a502fa>

Picture of two chess players: Taken by up and coming chess player and Math and Society enjoyer.

Hans Nieman Metal detector:

<https://www.tmz.com/2022/10/06/chess-grandmaster-hans-niemann-butt-scan-accuse-cheat-us-championship/>